

13.1.1. Fibonacci Sequence using Modules

AIM: Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Algorithm

Step 1: Start

Step 2: Input integer n

Step 3: If $n \leq 0$ Print "Please enter a positive integer" Stop

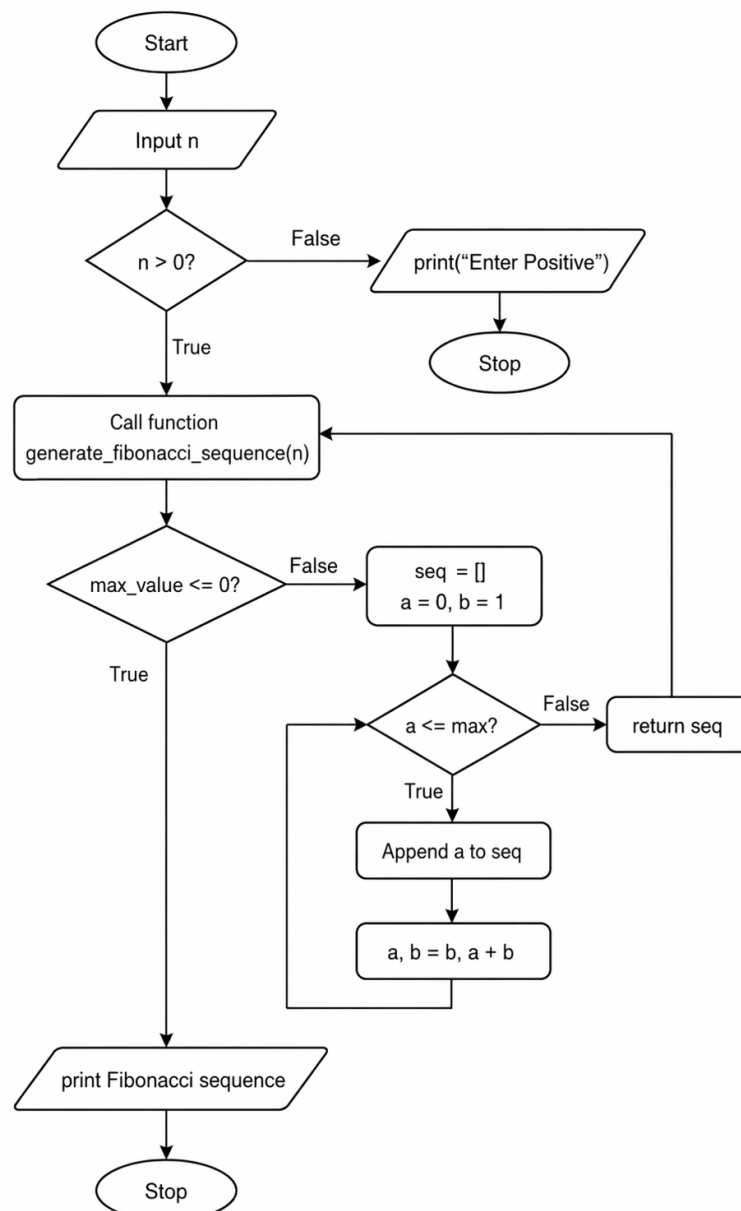
Step 4: Initialize: $a \leftarrow 0$ $b \leftarrow 1$ $seq \leftarrow$ empty list

Step 5: While $a \leq n$ do Add a to seq temp $\leftarrow a + b$ $a \leftarrow b$ $b \leftarrow$ temp

Step 6: Print all elements of seq

Step 7: Stop

Flowchart



EXECUTION

CODETANTRA

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13.1.1. Fibonacci Sequence using Modules05:45

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

- Create a module `fibonacci_module.py` with a function `generate_fibonacci_sequence(max_value)` that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create `main.py` that performs the following:

- Import the `fibonacci_module`.
- Read an integer `n` from user input.
- Validate if `n` is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer `n` representing the maximum value for Fibonacci sequence.

Output Format:

- If $n > 0$, print the Fibonacci sequence as space-separated numbers.
- If $n \leq 0$, print: "Please enter a positive integer".

Note:

- Two files are provided: `fibonacci_module.py` and `main.py`. The `main.py` file handles input and output. You need to complete both files based on the question requirements.

Sample Test Cases

main.py

1import fibonacci_module
2
3n = int(input())
4
5if n > 0:
6 fib_series = fibonacci_module.generate_fibonacci_sequence(n)
7 print(' '.join(map(str, fib_series)))
8else:
9 print("Please enter a positive integer")

Average time0.002 s1.75 msMaximum time0.004 s4.00 ms

4 out of 4 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 14.7msDebug

Expected output16
0 1 1 2 3 5 8
Actual output16
0 1 1 2 3 5 8

Test case 22ms

Test case 32ms

TerminalTest cases

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